

```

import pandas as pd
import numpy as np
from scipy import stats
import statsmodels.api as sm
import statsmodels.formula.api as smf
import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")

```

```

df = pd.read_csv("/content/titanic_cleaned.csv")
print("Dataset shape:", df.shape)
print(df.head())

```

Dataset shape: (775, 22)

	survived	pclass	sex	age	sibsp	parch	fare	class	who	\
0	0	3	male	22.0	1	0	7.2500	Third	man	
1	1	1	female	38.0	1	0	71.2833	First	woman	
2	1	3	female	26.0	0	0	7.9250	Third	woman	
3	1	1	female	35.0	1	0	53.1000	First	woman	
4	0	3	male	35.0	0	0	8.0500	Third	man	

	adult_male	...	sex_encoded	alive_encoded	embarked_C	embarked_Q	\
0	True	...	1	0	False	False	
1	False	...	0	1	True	False	
2	False	...	0	1	False	False	
3	False	...	0	1	False	False	
4	True	...	1	0	False	False	

	embarked_S	age_scaled	fare_scaled	family_size	age_group	\
0	True	0.371556	0.098747	2	Young Adult	
1	False	0.647039	0.970901	2	Adult	
2	True	0.440427	0.107941	1	Young Adult	
3	True	0.595386	0.723239	2	Young Adult	
4	True	0.595386	0.109644	1	Young Adult	

	fare_category
0	Low
1	High
2	Low
3	High
4	Low

[5 rows x 22 columns]

```
df.columns = df.columns.str.strip()
```

```

print("Shape of dataset:", df.shape)
print("\nColumns:")
print(df.columns)

```

```

print("\nDataset Information:")
df.info()

```

```

print("\nMissing Values:")
print(df.isnull().sum())

```

```
'family_size', 'age_group', 'fare_category'],  
dtype='object')
```

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 775 entries, 0 to 774
```

```
Data columns (total 22 columns):
```

#	Column	Non-Null Count	Dtype
0	survived	775 non-null	int64
1	pclass	775 non-null	int64
2	sex	775 non-null	object
3	age	775 non-null	float64
4	sibsp	775 non-null	int64
5	parch	775 non-null	int64
6	fare	775 non-null	float64
7	class	775 non-null	object
8	who	775 non-null	object
9	adult_male	775 non-null	bool
10	alive	775 non-null	object
11	alone	775 non-null	bool
12	sex_encoded	775 non-null	int64
13	alive_encoded	775 non-null	int64
14	embarked_C	775 non-null	bool
15	embarked_Q	775 non-null	bool
16	embarked_S	775 non-null	bool
17	age_scaled	775 non-null	float64
18	fare_scaled	775 non-null	float64
19	family_size	775 non-null	int64
20	age_group	775 non-null	object
21	fare_category	775 non-null	object

```
dtypes: bool(5), float64(4), int64(7), object(6)
```

```
memory usage: 106.8+ KB
```

Missing Values:

survived	0
pclass	0
sex	0
age	0
sibsp	0
parch	0
fare	0
class	0
who	0
adult_male	0
alive	0
alone	0
sex_encoded	0
alive_encoded	0
embarked_C	0
embarked_Q	0
embarked_S	0
age_scaled	0
fare_scaled	0
family_size	0
age_group	0
fare_category	0

```
dtype: int64
```

```
if 'age' in data.columns:  
    data['age'] = data['age'].fillna(data['age'].median())  
  
if 'fare' in data.columns:  
    data['fare'] = data['fare'].fillna(data['fare'].median())
```

```

if 'embarked' in data.columns:
    data['embarked'] = data['embarked'].fillna(
        data['embarked'].mode()[0]
    )

print("Missing values after cleaning:")
print(data.isnull().sum())

```

Missing values after cleaning:

```

survived      0
pclass        0
sex           0
age           0
sibsp         0
parch         0
fare          0
class         0
who           0
adult_male    0
alive         0
alone         0
sex_encoded   0
alive_encoded 0
embarked_C    0
embarked_Q    0
embarked_S    0
age_scaled    0
fare_scaled   0
family_size   0
age_group     0
fare_category 0
dtype: int64

```

```

numeric_columns = ['age', 'fare', 'sibsp', 'parch']

```

```

print("MEAN")
print(data[numeric_columns].mean())

print("\nMEDIAN")
print(data[numeric_columns].median())

print("\nMODE")
print(data[numeric_columns].mode().iloc[0])

print("\nVARIANCE")
print(data[numeric_columns].var())

print("\nSTANDARD DEVIATION")
print(data[numeric_columns].std())

print("\nCOMPLETE DESCRIPTIVE STATISTICS")
print(data[numeric_columns].describe())

```

```

MEAN
age      29.363123
fare     26.593137
sibsp    0.529032
parch    0.420645
dtype: float64

```

```

MEDIAN

```

```
age      28.0
fare     15.9
sibsp    0.0
parch    0.0
dtype: float64
```

MODE

```
age      28.00000
fare     73.41975
sibsp    0.00000
parch    0.00000
Name: 0, dtype: float64
```

VARIANCE

```
age      174.546333
fare     523.178676
sibsp     0.980745
parch     0.706550
dtype: float64
```

STANDARD DEVIATION

```
age      13.211598
fare     22.873099
sibsp     0.990326
parch     0.840565
dtype: float64
```

COMPLETE DESCRIPTIVE STATISTICS

	age	fare	sibsp	parch
count	775.000000	775.000000	775.000000	775.000000
mean	29.363123	26.593137	0.529032	0.420645
std	13.211598	22.873099	0.990326	0.840565
min	0.420000	0.000000	0.000000	0.000000
25%	21.000000	8.050000	0.000000	0.000000
50%	28.000000	15.900000	0.000000	0.000000
75%	36.000000	34.197900	1.000000	1.000000
max	58.500000	73.419750	8.000000	6.000000

```
print("Survival Count:")
print(data['survived'].value_counts())

plt.figure(figsize=(6, 4))

sns.countplot(x='survived', data=data)

plt.title("Titanic Survival Count")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Number of Passengers")

plt.show()
```

```
Survival Count:
survived
0      455
1      320
Name: count, dtype: int64
```

```
print("Survival by Gender:")

gender_survival = pd.crosstab(
    data['sex'],
    data['survived']
```

```

)

print(gender_survival)

plt.figure(figsize=(7, 5))

sns.countplot(
    x='sex',
    hue='survived',
    data=data
)

plt.title("Survival by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")

plt.show()

```

```

Survival by Gender:
survived    0    1
sex
female      76  216
male       379  104

```

```

correlation_columns = [
    'survived',
    'pclass',
    'age',
    'sibsp',
    'parch',
    'fare'
]

correlation_matrix = data[correlation_columns].corr()

print("Correlation Matrix:")
print(correlation_matrix)

```

```

Correlation Matrix:
      survived    pclass      age      sibsp      parch      fare
survived  1.000000 -0.331388 -0.075582 -0.037841  0.069864  0.306633
pclass   -0.331388  1.000000 -0.345319  0.087050  0.038998 -0.741242
age      -0.075582 -0.345319  1.000000 -0.284784 -0.187100  0.147240
sibsp    -0.037841  0.087050 -0.284784  1.000000  0.379535  0.263081
parch     0.069864  0.038998 -0.187100  0.379535  1.000000  0.238917
fare      0.306633 -0.741242  0.147240  0.263081  0.238917  1.000000

```

```

plt.figure(figsize=(8, 6))

sns.heatmap(
    correlation_matrix,
    annot=True,
    cmap='coolwarm',
    fmt='.2f'
)

plt.title("Titanic Correlation Matrix")
plt.show()

```

```
correlation, p_value = stats.pearsonr(
    data['age'],
    data['fare']
)

print("Pearson Correlation:", correlation)
print("P-value:", p_value)
```

Pearson Correlation: 0.1472404630896287
P-value: 3.874614238193193e-05

```
survived_age = data[data['survived'] == 1]['age']

not_survived_age = data[data['survived'] == 0]['age']

# Perform t-test
t_statistic, t_pvalue = stats.ttest_ind(
    survived_age,
    not_survived_age,
    equal_var=False
)

print("T-TEST RESULTS")
print("-----")
print("T-statistic:", t_statistic)
print("P-value:", t_pvalue)
```

T-TEST RESULTS

T-statistic: -2.080481156501386
P-value: 0.037869963180922256

```
alpha = 0.05

if t_pvalue < alpha:
    print("\nDecision: Reject H0")
    print("There is a significant difference in average age.")
else:
    print("\nDecision: Fail to Reject H0")
    print("There is no significant difference in average age.")
```

Decision: Reject H0
There is a significant difference in average age.

```
class_1 = data[data['pclass'] == 1]['fare']

class_2 = data[data['pclass'] == 2]['fare']

class_3 = data[data['pclass'] == 3]['fare']

# Perform ANOVA
f_statistic, anova_pvalue = stats.f_oneway(
    class_1,
    class_2,
    class_3
)
```

```
print("ANOVA RESULTS")
print("-----")
print("F-statistic:", f_statistic)
print("P-value:", anova_pvalue)
```

ANOVA RESULTS

```
-----
F-statistic: 573.9214015737475
P-value: 1.9042732700319744e-153
```

```
alpha = 0.05
```

```
if anova_pvalue < alpha:
    print("\nDecision: Reject H0")
    print("There is a significant difference in fare among passenger classes.")
else:
    print("\nDecision: Fail to Reject H0")
    print("There is no significant difference in fare among passenger classes.")
```

```
Decision: Reject H0
There is a significant difference in fare among passenger classes.
```

```
plt.figure(figsize=(8, 5))

sns.boxplot(
    x='pclass',
    y='fare',
    data=data
)

plt.title("Fare Distribution by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Fare")

plt.show()
```

```
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
import matplotlib.pyplot as plt
```

```
X = data[['age']]
```

```
y = data['fare']
```

```
regression_model = LinearRegression()
```

```
regression_model.fit(X, y)
```

```
y_pred = regression_model.predict(X)
s
print("Linear Regression Results")
print("-----")
```

```
print("Coefficient:", regression_model.coef_[0])
print("Intercept:", regression_model.intercept_)
```

```
r2 = r2_score(y, y_pred)
```

```
print("R² Score:", r2)
```

Linear Regression Results

Coefficient: 0.2549158425601123
Intercept: 19.10801177038479
R² Score: 0.021679753970848314

```
r2 = r2_score(y, y_pred)
```

```
print("R² Score:", r2)
```

R² Score: 0.021679753970848314

```
plt.figure(figsize=(8, 5))
```

```
plt.scatter(
    data['age'],
    data['fare'],
    alpha=0.5
)
```

```
plt.plot(
    data['age'],
    y_pred
)
```

```
plt.xlabel("Age")
plt.ylabel("Fare")
```

```
plt.title("Linear Regression: Age vs Fare")
```

```
plt.show()
```

```
X_stats = data[['age']]
```

```
X_stats = sm.add_constant(X_stats)
```

```
y_stats = data['fare']
```

```
model = sm.OLS(y_stats, X_stats).fit()
```

```
print(model.summary())
```

OLS Regression Results

```
=====
Dep. Variable:          fare    R-squared:                0.022
Model:                  OLS     Adj. R-squared:           0.020
Method:                 Least Squares    F-statistic:           17.13
Date:                   Wed, 02 Sep 2026    Prob (F-statistic):    3.87e-05
Time:                   12:35:31    Log-Likelihood:       -3516.4
```

```

No. Observations:      775   AIC:      7037.
Df Residuals:          773   BIC:      7046.
Df Model:              1
Covariance Type:      nonrobust
=====
              coef      std err          t      P>|t|      [0.025      0.975]
-----
const         19.1080        1.983        9.636      0.000        15.215        23.001
age            0.2549         0.062         4.139      0.000          0.134          0.376
=====
Omnibus:                 103.381   Durbin-Watson:                 2.037
Prob(Omnibus):            0.000   Jarque-Bera (JB):            145.082
Skew:                    1.054   Prob(JB):                     3.13e-32
Kurtosis:                 2.786   Cond. No.                      78.6
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly speci

```

print("\n===== FINAL CONCLUSION =====")

if t_pvalue < 0.05:
    print("1. T-Test: There is a statistically significant difference")
    print("    in the average age of survivors and non-survivors.")
else:
    print("1. T-Test: There is no statistically significant difference")
    print("    in the average age of survivors and non-survivors.")

if anova_pvalue < 0.05:
    print("2. ANOVA: Passenger class has a significant effect on fare.")
else:
    print("2. ANOVA: No significant difference in fare was found")
    print("    among passenger classes.")

print("3. Pearson Correlation between Age and Fare:",
      round(correlation, 4))

print("4. R² Score of Age vs Fare Regression:",
      round(r2, 4))

```

```

===== FINAL CONCLUSION =====
1. T-Test: There is a statistically significant difference
    in the average age of survivors and non-survivors.
2. ANOVA: Passenger class has a significant effect on fare.
3. Pearson Correlation between Age and Fare: 0.1472
4. R² Score of Age vs Fare Regression: 0.0217

```

Start coding or [generate](#) with AI.

